

SITONG LI

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Dear Anna, Bradford, and the Chicago Booth Hiring Committee,

I am writing to apply for the Research Professional position affiliated with the Center for Applied AI at Chicago Booth. I am currently a full-time Research Analyst at the Ross School of Business, University of Michigan, where I build and maintain ML and LLM-based data infrastructure for large-scale administrative research. The project you describe — web-scale processing of millions of academic papers, distributed web crawling on AWS, and real-time application of ML models and LLMs — is exactly the kind of infrastructure work I have been building toward, and the science of science framing is a research question I find genuinely compelling. I would be glad to bring that experience and enthusiasm to Chicago Booth for one to two years.

My technical background maps closely onto the infrastructure you have described. At Michigan Ross, I own two production pipelines: an entity resolution system integrating LLMs, fuzzy string matching, and word embeddings to link records across heterogeneous administrative datasets, and a computer vision pipeline that runs trained ML models against large-scale applicant data for demographic attribute extraction. Both systems run on out-of-memory datasets and required careful design for scalability, reliability, and maintainability. Prior to this role, I built a distributed web crawler to collect and parse 10,000+ full-text 10-K filings from SEC EDGAR for large-scale text analysis — the same combination of distributed web crawling, document parsing, and downstream ML application that sits at the core of your current project. I then deployed BERT-based classifiers across 500,000+ sentences from those filings using distributed compute (Spark, Databricks, AWS), and used GPT-4 for real-time structured information extraction from segmented text. For a separate project, I designed an agentic LLM pipeline to process 100,000+ regulatory documents across 1,000+ entities with zero manual intervention. Across all of these projects, the core engineering challenge was the same as yours: building systems that handle data at a scale that breaks naive approaches, and that remain maintainable as they grow.

Beyond the infrastructure work, I have broad experience with the analytical methods that sit downstream of data collection: regression models, panel data methods, BERT- and word-embedding-based NLP models, attention-based ML models, and XGBoost — all applied to real research questions at the publication or working-paper stage. I hold a Master of Science in Finance and Business Analytics from the University of Minnesota (GPA 3.97) and have worked as a research assistant on three concurrent faculty research projects, each involving independent ownership of the full data pipeline from collection through cleaning, modeling, and output delivery. I am comfortable operating in ambiguous, fast-moving research environments, communicating the status of long-running systems and surfacing problems early, and identifying opportunities for process improvement before they become bottlenecks.

I am deeply motivated by the academic trajectory this role supports. Working alongside faculty at Chicago Booth on a project at the frontier of AI and science of science research

— and doing so within the predoc program that has placed researchers at Chicago, Cornell, Harvard, MIT, and Stanford — is precisely the preparation I am seeking before a Ph.D. in economics or a related quantitative field. I am available to begin in July 2026 and committed to a two-year engagement. I would welcome the chance to discuss the project in more detail.

Sincerely,

Sitong Li